PRACTICE SHEET

1. Let A = {–1, 2, 5, 8}, B = {0, 1, 3, 6, 7} and R be the relation 'is one less than' from A to B, then how many elements will R contain?
(a) 2 (b) 3
(c) 5 (d) 9

2. If n(A) = 115, n(B) = 326, n(A – B) = 47, then what is n(A ∪B) equal to?
(a) 373 (b) 165
(c) 370 (d) 394

3. If P(A) denotes the power set of A and A is the void set, then what is number of elements in P{P{P{P(A)}}}?
(a) 0 (b) 1
(c) 4 (d) 16

4. If $N_a = \{ax \mid x \in N\}$, then what is $N_{12} \cap N_8$ equal to?
(a) N_{12} (b) N_{20}
(c) N_{24} (d) N_{48}

5. If $X = \{(4^n - 3n - 1) \mid n \in N\}$ and $Y = \{9(n - 1) \mid n \in N\}$, then $X \cup Y$ equals to
(a) X (b) Y
(c) N (d) A null set

6. Sets A and B have n element in common. How many

(b) R is symmetric and transitive
(c) R is transitive but not reflexive
(d) R is neither reflexive nor transitive

12. Let R be the relation defined on the set of natural number N as aRb; b∈N, if a divides b. Then, which one of the following is correct?
(a) R is reflexive only (b) R is symmetric only
(c) R is transitive only (d) R is reflexive and transitive

13. Consider the following statements
I. $\phi \in \{\phi\}$ (II) $\{\phi\} \subseteq \phi$
Which of the statements given above is/are Correct?
(a) I only (b) II only
(c) Both I and II (d) Neither I nor II

14. Which one of the following is correct?
(a) A ∪ P(A) = P (A) (b) A ∩ P(A) = A
(c) A – P(A) = A (d) P(A) – {A} = P(A)
Here, P(A) denotes the power set of a set A.

15. If A, B and C are three sets, such that A ∪ B = A ∪ C, then which one of the following is correct?
(a) A = B only (b) B = c only
(c) A = C only (d) A = B + C

16. If A, B and C are three sets, then A – (B – C) equal